ITLSE801 : SOFTWARE ENGINEERING

ACTIVITY 2: WRITING FURPS+ REQUIREMENTS AND A MINI SRS

Project: RP-Cloud Labs – Tool-Session Service

Group Members:

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Level: 8

Optional: IT

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of the RP-Cloud Labs Tool-Session Service. The service will allow students to request access to hosted learning tools through a web browser without installing the tools on their personal computers.

1.2 Scope

The Tool-Session Service will receive student requests, validate student accounts, check college software license capacity, and record session information. The system will help colleges manage access to learning tools such as programming environments, databases, and computer-aided design (CAD) applications.

The first version will focus on requesting sessions, checking license availability, and recording session details. It will not implement every feature of the complete RP-Cloud Labs platform.

2. General Description

2.1 Product Perspective

The Tool-Session Service is a component of the proposed RP-Cloud Labs platform. It will communicate with other platform components as necessary to validate accounts, check license capacity, and manage session information.

2.2 User Classes

- **Students:** Request access to learning tools.
- **Lecturers:** Use hosted tools for teaching and practical assessments.
- **College administrators:** Manage college access and software license information.
- **RP central IT staff:** Maintain and monitor the platform.

2.3 Assumptions and Constraints

- Users have access to a web browser and an internet connection.
- Students must have valid accounts to request sessions.
- License capacity information must be available and up to date.
- Access to a tool depends on the college's available license capacity.
- The system should support students using low-specification computers.

3. Functional Requirements

Functional requirements describe what the system must do.

FR-01: Request a Tool Session

Requirement: The system shall allow an authenticated student to select an available learning tool and submit a session request.

Acceptance criteria:

- An authenticated student can select a tool.
- The student can submit the request.
- The system returns a confirmation that the request was received.

FR-02: Validate Student Accounts

Requirement: The system shall verify that a student's account is active and authorized before processing a tool-session request.

Acceptance criteria:

- A request from an active, authorized student can proceed to the next validation step.
- A request from an inactive or unauthorized account is rejected.
- The system displays an appropriate message when a request is rejected.

FR-03: Check License Capacity

Requirement: The system shall check the selected college's available license capacity before allocating a tool session.

Acceptance criteria:

- A session can be allocated when the required license capacity is available and the request is valid.
- A request is not allocated when no license capacity remains.
- The system informs the student when capacity is unavailable.
- Simultaneous requests must not cause the system to allocate more sessions than the available capacity.

FR-04: Record Tool Sessions

Requirement: The system shall record each accepted tool-session request with a unique session ID, student ID, college ID, tool ID, request date and time, and session status.

Acceptance criteria:

- Every accepted request has a unique session ID.
- The required details are stored.
- Authorized users can retrieve the recorded session details.
- The session status is updated when its state changes.

Requirement Quality Check

Requirement	Clear	Complete	Consistent	Testable	Measurable
FR-01	Yes	Yes	Yes	Yes	Yes
FR-02	Yes	Yes	Yes	Yes	Yes
FR-03	Yes	Yes	Yes	Yes	Yes
FR-04	Yes	Yes	Yes	Yes	Yes

Validation note: These requirements have been reviewed for clarity, completeness, consistency, testability, and measurability. The group should confirm them against the stakeholder needs from Activity 1 and revise them if the review identifies any missing details.

4. Non-Functional Requirements Using FURPS+

FURPS+ is a framework for describing software quality requirements. It covers Functionality, Usability, Reliability, Performance, Supportability, and additional requirements such as security and portability.

F – Functionality

NFR-01: The system shall prevent session allocation when the student's account is inactive, the request is unauthorized, or the required license capacity is unavailable.

Measurement: Test each condition and confirm that no invalid session is allocated.

U – Usability

NFR-02: A student shall be able to submit a tool-session request using a web browser without needing command-line instructions.

Measurement: At least 9 out of 10 representative student testers shall complete the request task without assistance during usability testing.

R – Reliability

NFR-03: The system shall preserve accepted session records when a service component restarts unexpectedly.

Measurement: In 10 simulated restart tests, all previously accepted session records shall remain available and correct.

P – Performance

NFR-04: Under a test load of 100 concurrent session requests, at least 95% of valid requests shall receive an initial response within 2 seconds, excluding the time required to stream or launch the remote tool itself.

Measurement: Use a load-testing tool to record response times and calculate the percentage of requests completed within the target.

S – Supportability

NFR-05: The system shall generate timestamped application logs for session requests, validation failures, license-capacity decisions, and unexpected errors.

Measurement: Verify that each tested event produces a log entry containing the event type, timestamp, and relevant session or request identifier, without exposing passwords or other sensitive credentials.

Additional Requirement – Security

NFR-06: The system shall require authenticated access and enforce role-based authorization for student, lecturer, and administrator operations.

Measurement: Test authorized and unauthorized access attempts for each role and confirm that unauthorized operations are denied.

Additional Consideration: Scalability

Although security is selected as the additional requirement above, scalability is also important. The system should be designed so that its capacity can be increased as more colleges and users join. A future scalability requirement should define a measurable target after expected user numbers and workload have been agreed upon.

5. Interface Requirements

5.1 User Interface

The system shall provide a browser-based interface through which students can select a learning tool, submit a session request, and view the request status.

5.2 Software Interfaces

The Tool-Session Service may communicate with:

- An identity or account service to validate students.
- A license-management component to check available license capacity.
- A session-management component to record and update session details.

5.3 Hardware and Network Interfaces

Users will access the service through computers with a web browser and internet connectivity. The service will run on hosted infrastructure rather than requiring each student to install the learning tools locally.

5.4 Interface Feedback

The system shall display understandable messages for successful requests, invalid accounts, unavailable license capacity, and failed requests.

Repository: https://github.com/sarah20501/rp-cloud-labs

MVP milestone link: https://github.com/sarah20501/rp-cloud-labs/issues?q=is%3Aopen%20milestone%3AMVP: